

CareerHub — ATS Score Checker Engine

Scoring algorithm, role-benchmark presets, loading-state UX, and suggestion logic

This document defines exactly how the ATS Score Checker works end to end: what it checks, how the 0-100 score is computed, what happens for each of the seven target roles, what the loading screen shows while it works, and how suggestions are generated from the score breakdown. The algorithm design is grounded in documented, publicly reported behavior of real ATS platforms (Workday, Greenhouse, iCIMS, Lever) rather than invented rules.

Important framing, stated up front: no real ATS platform outputs a single literal "match percentage" to candidates -- that number is a UX convention every resume-checker tool uses as a readable proxy for a genuinely more complex parsing-and-ranking process. CareerHub's score is built the same way: a defensible, weighted model of the real factors ATS platforms are documented to check, not a copy of any vendor's proprietary formula. This should be stated in-product near the score, not just in this internal doc.

1. How Real ATS Platforms Work (Research Summary)

Modern ATS platforms run resumes through three phases: **parsing** (extracting structured fields from the document), **matching** (comparing those fields against job requirements), and **ranking** (ordering candidates for the recruiter). Failing parsing means never reaching matching, regardless of how qualified the candidate is.

Parsing -- what breaks it:

- Multi-column layouts and text boxes are the single most common failure -- parsers read left-to-right, top-to-bottom, so columns get read out of order or scrambled, and text inside text boxes/headers/footers is frequently skipped entirely.
- Tables used for page layout (not simple data tables) and nested tables cause the same scrambling.
- Non-standard section headings ("My Journey" instead of "Experience") prevent the parser from correctly categorizing content into database fields.
- Custom/downloaded fonts, icons used as bullets or contact-detail markers, and image-based content (skill bars, charts, photos) frequently render as blank, garbled, or dropped text.
- Non-text-selectable PDFs (i.e. a PDF that is actually a scanned image) cannot be parsed at all.

One formatting-guide source estimated formatting issues cause roughly 23% of ATS parsing failures; another estimated over 40% of resumes contain at least one element that causes some field to be dropped or misassigned. Exact figures vary by source and none are from a controlled study, but the direction is consistent enough across independent sources to treat formatting as a first-class, heavily-weighted scoring category rather than an afterthought.

Formatting rules with the most consistent support across sources:

- Single-column layout, standard section headers (Work Experience, Education, Skills), consistent date format throughout.
- Standard system fonts (Arial, Calibri, Helvetica, Georgia, Times New Roman), 10-12pt body text.
- No tables-for-layout, no text boxes, no headers/footers holding critical info, no icons/graphics/skill bars, simple round or dash bullets only.
- Text-selectable PDF (verify by trying to highlight text with the cursor) or DOCX.

Matching -- how keyword scoring actually works:

- Matching is not one algorithm: it combines exact-keyword search, Boolean filters, and (on modern systems like Workday, Greenhouse, and iCIMS) NLP-based semantic matching that can recognize related terms -- but exact, literal matches are still weighted more heavily than a synonym or paraphrase across every source reviewed.
- The exact job title from the posting is reported as the single highest-leverage keyword placement -- one industry data source (Jobscan) reported resumes containing the exact posted title were 10.6x more likely to lead to an interview.
- If five or more of a role's core required keywords are missing from a resume, match score is consistently reported as landing low regardless of the candidate's actual qualification -- this is a meaningful threshold to encode directly into the scoring model (see Section 3).
- Keyword stuffing is both detectable and counterproductive: sources converge on roughly 2-3 natural repetitions of an important term as healthy, with density-flagging behavior reported to activate above 4-5 repetitions of the same term.
- A commonly cited industry pass-through benchmark is roughly 70% match as "acceptable" for most roles, with competitive/senior roles effectively requiring 80%+.

Design implication: CareerHub's scoring model below mirrors this structure directly -- a heavily weighted formatting/parseability gate, exact-match-first keyword scoring with partial credit for close variants, an explicit job-title check, and a density check that penalizes both under- and over-optimization.

2. The Scoring Pipeline (End to End)

1. INTAKE -> resume text (from CareerHub-generated LaTeX/PDF, or user-uploaded file), selected target role (or pasted job description as an override)
2. PARSE -> extract raw text; for uploaded files, also extract layout signals (column structure, font list, embedded images/tables, text-selectability)
3. STRUCTURE -> detect standard sections (Summary, Experience, Education, Skills, Projects, Certifications) by header pattern matching
4. BENCHMARK -> load the target role's keyword/skill/title preset (Section 4), OR extract requirement keywords from a pasted job description
5. MATCH -> score resume content against the benchmark: exact match, partial/variant match, title match, keyword density check
6. QUALITY -> score achievement bullets for quantification and action-verb usage
7. AGGREGATE -> combine weighted category scores into the 0-100 total
8. SUGGEST -> generate specific, ranked suggestions from whichever categories scored below their target threshold
9. RENDER -> return score, category breakdown, and suggestions to the frontend

3. Weighted Scoring Model

Total score is out of 100, built from six weighted categories:

Category	Weight	What it checks
Parseability & Formatting	20	Single-column structure, standard section headers, consistent date format, no unparseable elements. CareerHub-generated resumes score near-full marks automatically here since every template is single-column and uses standard headers; uploaded resumes get the full heuristic check.
Job Title Alignment	10	Does the resume's most recent title, objective line, or summary reflect the target role's canonical title(s)? Weighted heavily on its own given the 10.6x interview-likelihood figure tied to exact title matches.
Keyword & Skill Coverage	30	Exact and partial matches against the role's must-have and nice-to-have keyword lists (Section 4), or against keywords extracted from a pasted job description.
Keyword Placement & Density	10	Are matched keywords distributed across Summary + Skills + Experience (not crammed into one section)? Penalizes both near-zero density and stuffing (5+ repeats of one term).
Achievement Quality	15	Ratio of experience/project bullets that start with a strong action verb AND contain a quantified result (number, percentage, currency, or scale word).
Section Completeness & Hygiene	15	Presence of expected sections for the role, consistent tense, no first-person pronouns, resume length appropriate for experience level (roughly 250-700 words for 1 page, more only if 2 pages is justified).

Score bands shown to the user:

Score	Label	Meaning shown to user
85-100	Strong match	Green. Resume is well-aligned; suggestions are minor polish only.

Score	Label	Meaning shown to user
70-84	Good, needs targeted fixes	Amber. Matches the commonly cited 70% industry pass-through benchmark; suggestions target the 2-3 lowest-scoring categories.
50-69	Significant gaps	Amber-red. Multiple categories below threshold; suggestions are prioritized by weight, highest-impact first.
Below 50	Not aligned to this role	Red. Usually driven by missing core keywords, a title mismatch, or a formatting/parseability failure -- lead the suggestions with whichever of these is the actual cause.

4. Keyword Matching Mechanics

Each keyword in a role's benchmark list is checked against the resume text in this order:

Match tier	Credit	Logic
1. Exact match	100% credit	Case-insensitive, word-boundary match of the exact term (e.g. "Spring Boot").
2. Known variant/synonym match	85% credit	A maintained synonym map catches common equivalents ("JS" / "JavaScript", "ML" / "Machine Learning", "CI/CD" / "Continuous Integration"). Mirrors documented ATS semantic-matching behavior, capped below exact-match credit since literal matches are still weighted higher industry-wide.
3. Stem/partial match	50% credit	Shared root only (e.g. resume has "managing" when benchmark wants "management") -- flagged in suggestions as "present but not in the expected form."
4. No match	0%, added to gap list	Feeds directly into the suggestion engine as a missing keyword, prioritized must-have before nice-to-have.

Density check (keyword placement category):

- For each matched keyword, count occurrences and note which sections it appears in.
- 0 occurrences after a partial/variant match elsewhere = still counts as present but flagged as weak.
- 1 occurrence in only one section = partial credit for placement (works, but not reinforced).
- 2-3 occurrences spread across Summary/Skills/Experience = full placement credit -- matches the commonly cited healthy range.
- 5+ occurrences of the same exact term = placement credit capped and a stuffing flag is raised in suggestions, since density-flagging is reported to activate around this point on modern systems.

5. Target Role Presets

Each preset supplies the benchmark the resume is scored against when the user selects that role instead of pasting a specific job description. Must-have keywords carry full weight in the coverage category; nice-to-have keywords add bonus coverage but are not penalized as gaps if missing. All keyword lists are standard, industry-recognized terms for each role -- not scraped or proprietary data -- and should be treated as a living list the team refines from real user outcomes over time.

Software Engineer / SDE (General)

Field	Preset content
Canonical titles	Software Engineer, Software Developer, SDE, Software Development Engineer
Must-have keywords	Data Structures, Algorithms, System Design, Git, REST API, Object-Oriented Programming, Debugging, Unit Testing, Agile, Problem Solving, CI/CD, Cloud
Nice-to-have keywords	Docker, Kubernetes, Microservices, AWS, Scalability, Design Patterns
Preferred action verbs	Built, Designed, Developed, Implemented, Optimized, Automated, Architected, Debugged, Deployed, Refactored
Summary benchmark	Leads with years of experience and core technical strengths (languages, systems), one measurable engineering outcome (performance, scale, or delivery speed), not soft skills.

Backend Developer

Field	Preset content
Canonical titles	Backend Developer, Backend Engineer, Server-Side Developer
Must-have keywords	REST API, Database Design, SQL, Node.js, Python, Java, Microservices, Authentication, API Integration, Server-Side Logic, Caching, Docker
Nice-to-have keywords	GraphQL, Message Queues, Redis, Kafka, Load Balancing, Kubernetes
Preferred action verbs	Built, Architected, Optimized, Integrated, Scaled, Deployed, Engineered, Automated
Summary benchmark	Emphasizes systems the candidate has built end-to-end and a concrete scale/performance number (requests/sec, latency reduction, uptime).

Java Developer

Field	Preset content
Canonical titles	Java Developer, Java Engineer, Backend Java Developer
Must-have keywords	Java, Spring Boot, Spring Framework, Hibernate, REST API, Multithreading, Collections, JVM, Maven, Microservices, SQL, JUnit
Nice-to-have keywords	Kafka, Docker, Kubernetes, AWS, Design Patterns, OOP
Preferred action verbs	Developed, Implemented, Optimized, Engineered, Built, Migrated, Debugged

Field	Preset content
Summary benchmark	Names the Java ecosystem stack explicitly (Spring, Hibernate, Maven) rather than just "Java" alone, since ecosystem terms are what most Java job descriptions actually filter on.

Marketing

Field	Preset content
Canonical titles	Marketing Executive, Marketing Associate, Digital Marketing Specialist, Marketing Specialist
Must-have keywords	SEO, Content Marketing, Social Media Marketing, Campaign Management, Google Analytics, Brand Strategy, Email Marketing, Market Research, A/B Testing, ROI, Lead Generation
Nice-to-have keywords	Google Ads, Meta Ads Manager, CRM, Marketing Automation, Copywriting
Preferred action verbs	Launched, Grew, Increased, Managed, Executed, Optimized, Drove, Generated
Summary benchmark	Leads with a quantified growth or campaign-performance result (traffic %, conversion rate, audience growth) rather than a list of responsibilities.

Sales

Field	Preset content
Canonical titles	Sales Executive, Sales Associate, Business Development Executive, Account Executive
Must-have keywords	Lead Generation, Client Relationship Management, Negotiation, Sales Pipeline, CRM, Cold Calling, Quota Attainment, Prospecting, Closing Deals, Revenue Growth
Nice-to-have keywords	Salesforce, B2B Sales, Account Management, Upselling
Preferred action verbs	Closed, Generated, Exceeded, Negotiated, Managed, Grew, Secured, Achieved
Summary benchmark	Leads with quota attainment or revenue figures explicitly (e.g. "achieved 120% of quota") -- this is the single highest-signal line for sales resumes across every source reviewed.

Customer Care / Support

Field	Preset content
Canonical titles	Customer Support Executive, Customer Service Representative, Customer Care Executive, Client Support Associate
Must-have keywords	Customer Support, Query Resolution, SLA Compliance, CRM, Escalation Handling, Communication Skills, Ticketing System, Customer Satisfaction, Chat Support, Email Support
Nice-to-have keywords	Zendesk, Freshdesk, Salesforce Service Cloud, Multichannel Support
Preferred action verbs	Resolved, Handled, Assisted, Managed, Responded, Improved, Maintained
Summary benchmark	Leads with a resolution-rate, response-time, or CSAT figure if available; otherwise volume handled (e.g. tickets/day) is the strongest available signal.

Default / General

Field	Preset content
Canonical titles	
Must-have keywords	Communication, Teamwork, Problem Solving, Time Management, Documentation, Attention to Detail
Nice-to-have keywords	Project Management, Leadership, Adaptability, Microsoft Office
Preferred action verbs	Managed, Coordinated, Assisted, Supported, Organized, Completed
Summary benchmark	Used when no specific role is selected, or for early-career profiles without a clear target yet -- scoring leans more heavily on formatting, achievement quantification, and section completeness than on role-specific keyword coverage, since there is no specific benchmark to match against.

When a user pastes an actual job description instead of selecting a preset role, the same weighting model applies, but the must-have/nice-to-have keyword lists are extracted from that job description directly: frequency-ranked noun phrases after stopword removal, with the first 12-15 unique terms treated as must-have and the next 8-10 as nice-to-have. This is the more accurate mode when available, since it reflects the exact posting rather than a general role average.

6. Loading-State UX

Every message the loader shows corresponds to a real pipeline step actually executing at that moment -- never a fixed decorative timer disconnected from real work. If a step legitimately takes under 300ms, give it a small minimum display floor so the sequence still reads clearly to the user; never show a stage that isn't backed by real computation happening at that instant.

Loader message	Backend step	What's actually happening
Reading your resume...	Pipeline step 2 (PARSE)	Text extraction; for uploaded files, layout-signal extraction (columns, fonts, tables, text-selectability).
Checking file structure...	Pipeline step 2-3 (PARSE/STRUCTURE)	Formatting/parseability heuristics; section-header detection.
Comparing against [Role] benchmark...	Pipeline step 4-5 (BENCHMARK/MATCH)	Loads the selected role preset (or extracts JD keywords) and runs keyword matching.
Finding gaps...	Pipeline step 5 (MATCH)	Computes the missing/weak keyword list feeding into suggestions.
Scoring your achievements...	Pipeline step 6 (QUALITY)	Bullet-level action-verb and quantification analysis.
Finalizing your score...	Pipeline step 7-8 (AGGREGATE/SUGGEST)	Weighted aggregation and suggestion generation.

Rule: the loader sequence length should scale with actual processing time. A CareerHub-generated resume (known-clean structure) can skip most of the formatting-check display time since that check is nearly instant against a known template; an uploaded resume genuinely takes longer at that step, and the UI should let it run that long rather than compressing it artificially.

7. Suggestion Generation Rules

Suggestions are generated per category, ranked by that category's weight (highest-weight gaps surface first), and always name the specific missing item -- never a generic tip.

Trigger	Generated suggestion pattern
Parseability below 80%	"[Detected issue] is likely to cause parsing errors on most ATS platforms. [Specific fix]." e.g. "This resume uses a two-column layout, which can cause your Experience section to be read out of order. Switch to a single-column format."
Title alignment below 60%	"Your most recent title doesn't match how this role is usually posted. Consider using '[canonical title]' if it honestly reflects your role."
Keyword coverage below threshold	"This resume is missing [N] core keywords for [Role]: [list up to 5 highest-priority missing must-have terms]. Add ones that are true of your actual experience."
Density/placement weak	"'[Keyword]' appears once, only in Skills. Reinforcing it in a bullet under Experience would strengthen the match."
Density stuffing flagged	"'[Keyword]' appears [N] times, which may read as keyword stuffing. Consider trimming to 2-3 natural mentions."
Achievement quality below 60%	"[N] of [Total] experience bullets have no measurable result. Add a number, percentage, or scale where honestly possible, e.g. 'Reduced X by 20%' rather than 'Responsible for X.'"

Trigger	Generated suggestion pattern
Section completeness gap	"No [Section name] section detected. Most [Role] resumes include this -- add it if applicable."

Hard rule: suggestions may only reference facts already present in the user's resume text or profile data. The engine may suggest adding a keyword the user's real experience plausibly supports, phrased as a prompt for the user to confirm and add themselves -- it must never insert a skill, tool, or achievement into the resume on the user's behalf without their action.

Sources synthesized for this document include current (2026) guidance from Jobscan, Enhancv, and several independent ATS-focused publications on parsing behavior, formatting rules, and keyword-match mechanics across Workday, Greenhouse, iCIMS, Lever, and Taleo. Figures cited (10.6x, 70% benchmark, 23%/40% formatting-failure estimates) are as reported by those sources, not independently verified by CareerHub, and should be treated as directionally reliable industry figures rather than precise statistics.